Mun Hyeok Chang – Curriculum Vitae

Biorobotics Laboratory
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Research Interests

- Prosthetic/Robotic hand
- Tendon-driven mechanism
- Under-actuated mechanism
- Variable transmission mechanism
- Biomimetic robotic system
- Soft robotic system

Experience

Jan. 2016 - Jan. 2017	**Associate** Quality Assurance Team, Daejeon Plant, Hanwha Corporation
Jan. 2009 - Feb. 2011	**Aircraft engine mechanic** Republic of Korea Air Force

Education

Sep. 2017 - present	**Ph.D. candidate in Mechanical Engineering** Seoul National University, Seoul, Korea Advisor: Prof. Kyu-Jin Cho
Mar. 2007 - Feb. 2016	**B.S. in Mechanical Engineering** Seoul National University, Seoul, Korea

PUBLICATIONS

International Journals

1. **Mun Hyeok Chang**, Inchul Jung, Kang Woo Seo, Jonghoo Park, Hyungmin Choi and Kyu-Jin Cho, "Posture-dependent variable transmission mechanism for prosthetic hand inspired by human grasping characteristics", in Intelligent Service Robotics (I.F. 4.3), Vol. 17, pp. 389–399, Apr. 2024, doi: 10.1007/s11370-024-00516-7.
2. Jonghoo Park, Mun Hyeok Chang, Inchul Jung, Haemin Lee, and Kyu-Jin Cho, "3D Printing in the Design and Fabrication of Anthropomorphic Hands: A Review", in Advanced Intelligent Systems, Vol. 6, 2300607, Apr. 2024, doi: 10.1002/aisy.202300607.
3. **Mun Hyeok Chang**, **Dong Hyun Kim**, Sang-Hun Kim, Yechan Lee, Seongyun Cho, Hyung-Soon Park and Kyu-Jin Cho, "Anthropomorphic Prosthetic Hand Inspired by Efficient Swing Mechanics for Sports Activities", in ***IEEE/ASME Transactions on Mechatronics*** **(I.F 5.673, Top 5%)**, Early Access, May. 2021, doi: 10.1109/TMECH.2021.3084311.

Patents

4. Kyu-Jin Cho, Sang-Hun Kim, **Mun Hyeok Chang**, “WIRE DRIVEN SPORTS PROSTHETIC HAND”, 10-2020-0074631, KR.

Refereed Conference Papers

5. **Mun Hyeok Chang**, Su Hwan Chae, Hye Ju Yoo, Sang-Hun Kim, Woongbae Kim, Kyu-Jin Cho, Loco-sheet: Morphing Inchworm Robot Across Rough-terrain, IEEE International Conference on Soft Robotics (RoboSoft), Seoul, Korea, 2019, ***Finalist***.

Research Experience

In Biorobotics Laboratory, Seoul National University, Seoul, Korea

2025 – 2026

Compact Dual-Motor Variable Transmission Utilizing Mutilated Gears

- Developing a compact dual-motor variable transmission mechanism for tendon-driven robotic hand actuation.
- Designed a transmission concept that combines two motor outputs with different reduction ratios without using a separate differential drive or brake mechanism.
- Utilized mutilated gears to selectively switch between high-speed tendon winding and high-force actuation modes.

2023 – 2026

Anthropomorphic Prosthetic Hand with Embedded Actuation for Precision and Tool-Oriented Grasping

- Designed an anthropomorphic prosthetic hand for diagonal power grip, tip pinch, and lateral pinch.
- Developed a geometric optimization method for active palm-joint placement to improve tool-oriented diagonal grasp stability.
- Proposed a passive thumb–finger actuation-order mechanism that switches between tip-pinch and lateral-pinch sequences using spring–magnet thresholds.
- Integrated dual-motor tendon actuation modules to combine fast grasp closure and increased tendon force in a compact prosthetic hand.
- Conducted modeling, prototyping, pull-out tests, motion capture experiments, fingertip force measurements, and functional demonstrations.

2022 – 2024

Posture-Dependent Variable Transmission Mechanism for Prosthetic Hands

- Developed a posture-dependent variable transmission mechanism using a spiral-shaped spool and a series elastic compression spring for tendon-driven prosthetic fingers.
- Designed the mechanism to enable fast pre-contact finger motion and higher post-contact tendon tension according to finger posture.
- Conducted modeling and parametric studies of spool geometry and spring characteristics to analyze the speed–force trade-off.
- Fabricated a 70 g PDVT-equipped prosthetic finger using 3D printing and metal components.
- Evaluated grasping speed, fingertip force, and durability, demonstrating up to 3.7× faster grasping speed and up to 5× stronger grip force than conventional rigid spool mechanisms.

2020 – 2025

High-Power Prosthetic Hand with Enhanced Durability

- Designed a high-power prosthetic hand targeting 1-second grasping and fingertip force greater than 45 N within a human-hand-sized form factor.
- Developed tendon routing and a moving-pulley differential actuation mechanism for integrating high-power motors into a compact prosthetic hand.
- Designed and analyzed moving-pulley components using FEM simulations to satisfy high fingertip-force requirements.
- Selected tendon materials and designed durability experiments for long-term tendon-driven actuation.
- Conducted repeated actuation tests, validating operation over more than 300,000 cycles.

2017 – 2021

Anthropomorphic Prosthetic Hand Inspired by Efficient Swing Mechanics for Sports Activities

- Designed an anthropomorphic prosthetic hand for diagonal power squeeze grip in sports activities.
- Developed articulated fourth and fifth CMC joints and 2-DOF MCP joints to reproduce palm concavity and improve diagonal club grasping.
- Designed FDP/FDS-inspired tendon routing for adaptive finger motion and increased grasp stability against external forces.
- Implemented a differential tendon-driven actuation system to drive multiple finger tendons with fewer actuators.
- Conducted pressure distribution and impact-resistance experiments, demonstrating increased ulnar-

side finger participation and improved grasp resilience compared with a non-articulated palm design.

2018 – 2019 **Loco-sheet: Morphing Inchworm Robot Across Rough-terrain**
- Developed a flexible sheet-based inchworm robot for rough and constrained environments.
- Proposed an S-shape locomotion mode that uses the bending stiffness of a PVC sheet to lift the robot's center of mass and climb stairs.
- Designed an omega-shape two-anchor crawling mode with anisotropic friction wheels for low-clearance locomotion.
- Implemented a motor-tendon actuation system for reversible morphing between S-shape and omega-shape locomotion modes.
- Fabricated and tested a fully untethered robot, demonstrating 90 mm stair climbing, 72 mm low-gap traversal, and flat-ground locomotion.

2017 – 2017 **Development of Biomimetic Bionic Hand Mechanism**
- Developed a tilted rolling contact joint for a compact biomimetic prosthetic finger mechanism.
- Designed tendon routing paths for finger actuation and contributed to joint design, mechanical implementation, and prototyping.

Technical Skills

Design & Manufacturing, Embedded systems, Analysis

- *Various prototyping experiences* (Prosthetic hand design, Soft robot design, variable transmission, experimental setups)
- *Controller design and simulation* (MATLAB, LabVIEW, etc.)
- *Prototyping control system* (Arduino)
- *Analysis* (MATLAB, SOLIDWORKS FEM)
- *CAD design* (SOLIDWORKS)
- *Manufacturing* (CNC milling, laser cutter, 3D printing, etc.)
- *PCB design and layout* (Fusion)

Scholarship

Mar. 2020 - Jun. 2020 Mar. 2019 - Jun. 2019 Sep. 2018 - Dec. 2018	**Brain Korea 21 Research Scholarship** *Funded by National Research Foundation of Korea*
Mar. 2018 - Jun. 2018	**Lecture & Research Scholarship** *Funded by Seoul National University*
Mar. 2012 - Jun. 2012 Mar. 2008 - Jun. 2008 Sep. 2007 - Dec. 2007 Mar. 2007 - Jun. 2007	**National Science & Technology Scholarship** *Funded by Korea Student Aid Foundation*

Honors and Awards

Apr. 2019	Conference Paper *Finalist*, 2019 IEEE International Conference on Soft Robotics (RoboSoft)
Apr. 2019	First place award, RoboSoft 2019 competition-Terrestrial Race, IEEE International Conference on Soft Robotics (RoboSoft)
Apr. 2018	Second place award, RoboSoft 2018 competition-Locomotion, IEEE International Conference on Soft Robotics (RoboSoft)
Jun. 2015	Grand prize, Creative design competition, Seoul National University
Jun. 2014	Gold award, 6th design contest for the Other 90%, Sharing and Technologies, Inc.

Teaching Experience

Sep. 2021 - Dec. 2021 Sep. 2022 - Dec. 2022 Sep. 2023 - Dec. 2023	**Adjunct faculty** Engineering Design *Ewha Womans University*
Sep. 2022 - Dec. 2022	**Adjunct faculty** Biomedical Signal Processing *Ewha Womans University*
Aug. 2025 – present, May. 2023 - July. 2023	**UROP Tutoring** Mentored two undergraduate students through the Undergraduate Research Opportunities Program.
Sep. 2019 - Dec. 2019 Sep. 2021 - Dec. 2021	**Teaching Assistant** Dynamics *Seoul National University*
Mar. 2018 - Nov. 2019	**Tutoring** CAD Design and 3D printing workshop *Haedong Idea Factory, Seoul National University*